



Direct Quantum Communication Laboratory · Yerevan, Armenia

What We Mean by “Quantum Communication”

An Illustrated Primer for the Nairi Quantum Team – Version 2

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Abstract

Even experienced scientists conflate *quantum communication* with quantum computing, quantum algorithms, or quantum key distribution. This note fixes the vocabulary and the physics. We start from the qubit, place every protocol on one family tree, explain each with a diagram and its essential mathematics, and separate what lives in the *physical* quantum domain from what is *classical* mathematics. We treat the physical layer—the electromagnetic spectrum, carrier, channel, encoding, source, and the two receiver architectures (DV and CV)—and build the bridge from classical software-defined radio (I/Q) to the continuous-variable quantum receiver (X/P). It closes by stating precisely what Nairi Quantum builds.

New in Version 2. Added: the electromagnetic-spectrum map (§8); the I/Q $\leftrightarrow$ X/P bridge with the Heisenberg derivation, coherent-state “blob,” and CV-QKD (§12); explicit proofs and pictures for *no-cloning* (§1) and *entanglement* (§5); a fuller *teleportation* walk-through (§6); *decoherence and quantum coding* in detail (§7); and expanded *encodings* (polarisation, time-bin, OAM) and *DV vs. CV* physics (§10,§11). Boxes in blue are the maths; boxes in gold are the plain-language conclusions.

Common misconceptions — read this first

You may hear...	The reality
“Quantum communication = quantum computers.”	Different pillar. Computers <i>process</i> qubits; communication <i>moves</i> them (§2).
“It’s just quantum mechanics / physics.”	Physics is the toolbox; communication is the application.
“Quantum communication just means QKD.”	QKD is one branch; we do <i>QSDC</i> (§4).
“It’s all maths, not real photons.”	No — genuine single photons (or coherent light) are prepared, sent, and measured (§1).
“Radio, microwave and light are different things.”	All are <i>electromagnetic waves</i> ; they differ only in wavelength (§8).
“The secret key is transmitted over the channel.”	The key is <i>never sent</i> — it emerges at both ends (§3).
“Entanglement alone sends a message.”	No — it is a shared <i>resource</i> ; a message needs classical bits too (§5,§6).

1. First, the qubit — the atom of everything below

A classical bit is 0 or 1. A **qubit** can be both at once—a normalised superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

When you *measure* it in the $\{|0\rangle, |1\rangle\}$ basis you get 0 with probability $|\alpha|^2$ or 1 with probability $|\beta|^2$, and the state *collapses* to whatever you got. Geometrically a qubit is a point on the *Bloch sphere* (Fig. 1):

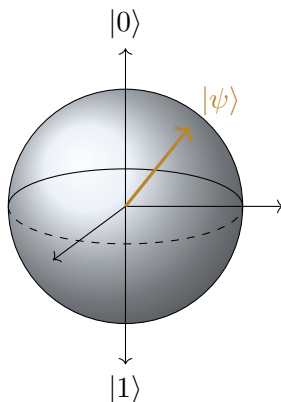


Figure 1: The Bloch sphere: a qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is a point on the unit sphere, with $|0\rangle$ and $|1\rangle$ at the poles.

What “both 0 and 1 at once” really means (a common sticking point). It is *not* “the bit is secretly 0 or 1 and we don’t know which” (that is classical ignorance). A superposition is a genuine third situation: the qubit has *no definite value* until measured, yet it behaves predictably in ways ignorance cannot explain. Example: prepare the diagonal state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Measured in the $\{|0\rangle, |1\rangle\}$ basis it gives 0 or 1 with probability $\frac{1}{2}$ each—looks random. But measured in the *diagonal* basis $\{|+\rangle, |-\rangle\}$ it gives $|+\rangle$ with certainty. A coin that were “really” 0 or 1 could never do that. The superposition is physically real; measurement is what forces a single outcome and erases the rest.

Two facts make quantum *communication* secure:

- **No-cloning theorem.** An unknown qubit cannot be copied. An eavesdropper cannot silently duplicate a photon in flight.
- **Measurement disturbs.** Any measurement in the wrong basis randomises the state, so eavesdropping leaves *statistical fingerprints*. This is the security engine.

No-cloning, proved (it is a mathematical fact, not an engineering limitation). Suppose a “universal copier” unitary U existed with $U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$ for *every* $|\psi\rangle$. Apply it to the basis states: $U(|0\rangle|0\rangle) = |0\rangle|0\rangle$ and $U(|1\rangle|0\rangle) = |1\rangle|1\rangle$. Now feed it $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Because quantum evolution is *linear*,

$$U(|\psi\rangle|0\rangle) = \frac{1}{\sqrt{2}}(U|0\rangle|0\rangle + U|1\rangle|0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

But genuine cloning would require $|\psi\rangle|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. These two results are different, so no such U exists. *Physically*: copying an unknown state would violate the linearity of quantum mechanics. This is why an eavesdropper cannot tap a quantum line the way she taps a fibre carrying classical light.

“Eavesdropping leaves statistical fingerprints”—made concrete. In BB84 (§3) an eavesdropper who intercepts a photon must guess Alice’s basis. She is wrong half the time, and each wrong guess randomises the bit Bob later receives. The result: she injects an error in about 25% of the sifted bits. Alice and Bob sacrifice a random sample and *estimate this error rate* (the QBER). No single bit proves anything—individual outcomes are random—but the *error statistics* over many bits betray her. Above a threshold QBER, they abort. (In CV systems the analogue is *excess noise* above the shot-noise floor, §12.)

Physical vs. mathematical — keep this. A qubit is an abstract object $\alpha|0\rangle + \beta|1\rangle$. To *communicate*, it must be *carried by real physics*—almost always a photon, with $|0\rangle, |1\rangle$ mapped onto a physical property (polarisation, time-bin, phase, or twist; §10). So quantum communication is *not* “maths only.” What stays classical is the ordinary digital processing around it (basis comparison, error correction, and encrypting the payload).

2. The family tree: QKD and QSDC are members, not synonyms

Quantum communication is an umbrella (Fig. 2). Its branches differ by *what actually travels through the quantum channel*:

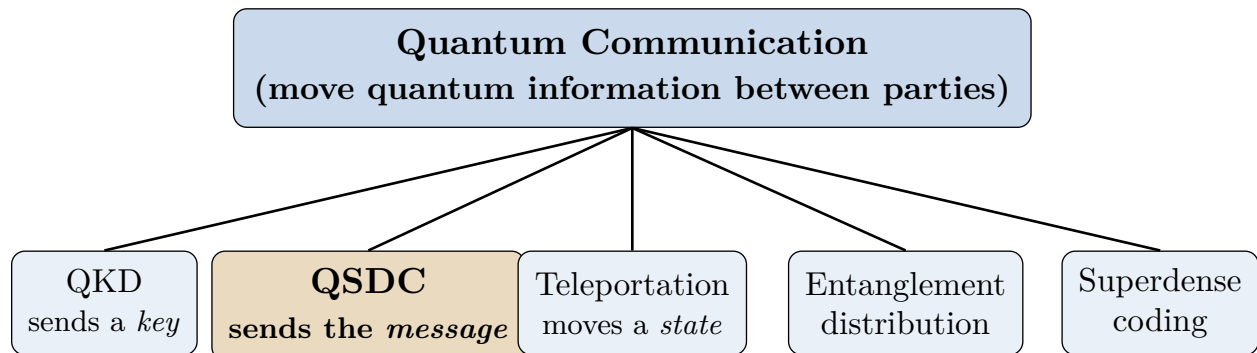


Figure 2: The quantum-communication family tree. Branches differ by what travels the quantum channel; QSDC (gold) — sending the message itself — is ours.

QKD hands you a key; QSDC delivers the secret message itself. The gold node is ours. Beyond these *protocols*, there are protocol *families* that harden or extend them: measurement-

device-independent QKD (MDI-QKD, immune to detector attacks), twin-field QKD (TF-QKD, longer reach), and distributed-phase-reference schemes (DPS, COW). They are variations on the same ideas, not a separate physics.

3. QKD — how it actually works (BB84)

QKD does *not* send your message. It lets Alice and Bob *grow the same random secret key* at their two ends, then that key encrypts the message with ordinary classical cryptography. The classic protocol is **BB84** (Fig. 3), which is a *discrete-variable* (DV, single-photon) protocol. Alice encodes each random bit on a single photon using a *randomly chosen basis*:

Z-basis (rectilinear): $|0\rangle = \uparrow$, $|1\rangle = \leftrightarrow$; X-basis (diagonal): $|+\rangle = \nearrow$, $|-\rangle = \nwarrow$.

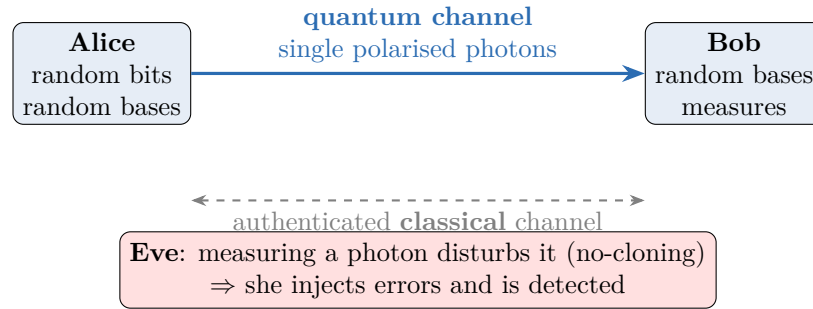


Figure 3: BB84 QKD. Single polarised photons cross the quantum channel; an authenticated classical channel handles basis announcement, sifting and error correction; Eve is revealed by the disturbance she must cause.

The steps. (1) Alice sends photons. (2) Bob measures each in his own random basis. (3) Over a *public but authenticated* classical channel they announce *which bases* they used (never the bit values) and keep only matched events—*sifting*. (4) They sacrifice a few bits to estimate the error rate; too many errors $\Rightarrow$ Eve $\Rightarrow$ abort. (5) Error correction + privacy amplification distil an identical secret key k . The message is then encrypted classically, e.g. one-time pad $c_i = m_i \oplus k_i$, or k seeds AES.

Worked example (sifting). Alice sends 8 photons; Bob measures in his own random bases; they keep only the positions where the bases happened to match.

Alice's bit	1	0	1	1	0	0	1	0
Alice's basis	Z	X	X	Z	Z	X	Z	X
Photon sent	$\leftrightarrow$	$\nearrow$	$\nwarrow$	$\leftrightarrow$	$\uparrow$	$\nearrow$	$\leftrightarrow$	$\nearrow$
Bob's basis	Z	Z	X	X	Z	X	Z	Z
Bases match?	✓	×	✓	×	✓	✓	✓	×
Sifted key	1		1		0	0	1	

Only the matched columns survive, giving both parties the *identical* sifted key 11001. On average *half* the photons survive sifting; a sacrificed subset is then checked for errors (Eve), and the rest is compressed to the final secret key.

Answering the encoding question directly. Yes—in BB84 each key bit is *encoded on a single photon* (typically its polarisation), and that photon is *physically transmitted over the optical link, which we call the quantum channel*. Your understanding is correct: one valid recipe is “random bit + random basis → prepare a photon → send it down the fibre/free-space.” BB84 is DV (single photons), *not* CV. The *key* itself is never transmitted; it is computed at both ends from the surviving measurements. The *classical* channel carries only non-secret bookkeeping.

Answering two more questions directly. “*Is the encryption done in the quantum domain / with qubits?*” **No.** The *key* is established using physical photons (the real quantum domain); encrypting the actual *message* with that key is classical maths (XOR / AES). Quantum physics protects the *key*; classical crypto protects the *payload*.

“*If the secret goes on the classical channel, does everything else go on the quantum channel?*” **Careful—the key is never transmitted at all.** Random qubits travel the quantum channel; Alice and Bob each *compute* the key locally from their own data. The classical channel carries only non-secret bookkeeping (which bases, error-correction syndromes)—authenticated but public. The final key then encrypts the message on a normal classical link. So there are **three roles**: the *quantum channel* = raw qubits; the *classical channel* = public reconciliation; the *key* = emerges at both ends, sent nowhere.

4. QSDC — sending the message itself

QSDC removes the “key first” step: the *confidential message* is carried *through the quantum channel*, with security from quantum mechanics rather than a pre-shared key—no key distribution, no ciphertext, information-theoretically secure in principle [4, 5, 6]. Practically it uses a *two-step / block* structure: a batch of photons is sent and its security *checked* on a sample; only if the channel is proven safe does Alice encode the real message on the remaining photons. It is fully in the *physical* photon domain—exactly Nairi Quantum’s target.

5. Entanglement — the shared quantum resource

Entanglement is a correlation with no classical analogue. Two qubits can be in a *Bell state*

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

which *cannot* be written as a product of two independent qubits. Measure either qubit and you get 0 or 1 with probability $\frac{1}{2}$; but the two outcomes are always *perfectly correlated*, no matter how far apart.

Entanglement in one line of algebra (and no-cloning). *Separable (not entangled):* if the two qubits were independent, the joint state would factor,

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle.$$

To match the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ we would need the $|01\rangle$ and $|10\rangle$ terms to vanish ($ad = 0$ and $bc = 0$) while $ac \neq 0$ and $bd \neq 0$ —a flat contradiction ($ad = 0$ forces $a = 0$ or $d = 0$, killing ac or bd). So **no product of two independent qubits equals $|\Phi^+\rangle$: it is entangled.** A separable example, for contrast, is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ —this *does* factor, so it is *not* entangled.

No-cloning, restated. Suppose a copier U gave $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$. Linearity on $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ yields $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, but true cloning demands $|\psi\rangle|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ —different states, so no such U exists (DV). **The CV version is the same idea:** a coherent state cannot be copied either, because a perfect copier would let you measure X on one copy and P on the other and beat $\Delta X \Delta P \geq \frac{1}{4}$. No-cloning holds for *both* DV and CV.

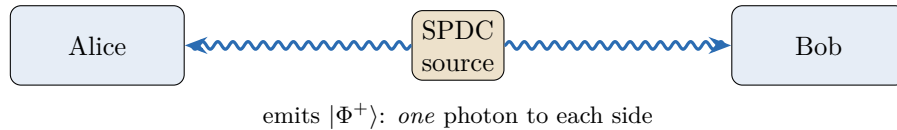


Figure 4: An entangled-pair (SPDC) source emits the Bell state $|\Phi^+\rangle$, sending *one* photon to Alice and *one* to Bob.

Correcting a natural misconception (Fig. 4). The source emits a *pair*; **one photon goes to Alice and the other to Bob**—*not* both photons to Bob. When Bob measures *his one* photon and gets, say, 0, he immediately knows Alice’s photon will also read 0 (for $|\Phi^+\rangle$). But this by itself **sends no message**: each result is individually random, and to *use* the correlation the parties must also exchange ordinary classical information. So entanglement is a shared *resource / fuel*, not communication on its own. It powers teleportation, entanglement-based QKD (Ekert), superdense coding, and the future quantum internet. “Entanglement distribution” just means reliably delivering these pairs over distance (the job of quantum repeaters).

What is SPDC? *Spontaneous Parametric Down-Conversion.* A strong “pump” laser photon is sent into a nonlinear crystal; occasionally one pump photon splits into *two* lower-energy “daughter” photons. Energy and momentum must be conserved, which forces the pair to be *correlated* (entangled) in polarisation, time, or direction. It is the workhorse source of entangled photon pairs.

Entanglement in CV, too (not DV-only). The CV version of a Bell pair is a *two-mode-squeezed* state (an “EPR” pair of light beams): the beams’ quadratures are correlated beyond any classical limit, $X_1 \approx X_2$ and $P_1 \approx -P_2$, tighter than $\Delta X \Delta P \geq \frac{1}{4}$ would allow for two independent beams (Fig. 5, right panel). It is produced by a nonlinear crystal (parametric down-conversion) and is the resource for *CV teleportation* and entanglement-based CV-QKD. Alongside it, a *squeezed* state (left/middle) shows the other non-classical trick: squeeze the noise blob into an ellipse, quieter than the vacuum in one quadrature.

Non-classical light: squeezing beats the vacuum floor; entanglement correlates two beams

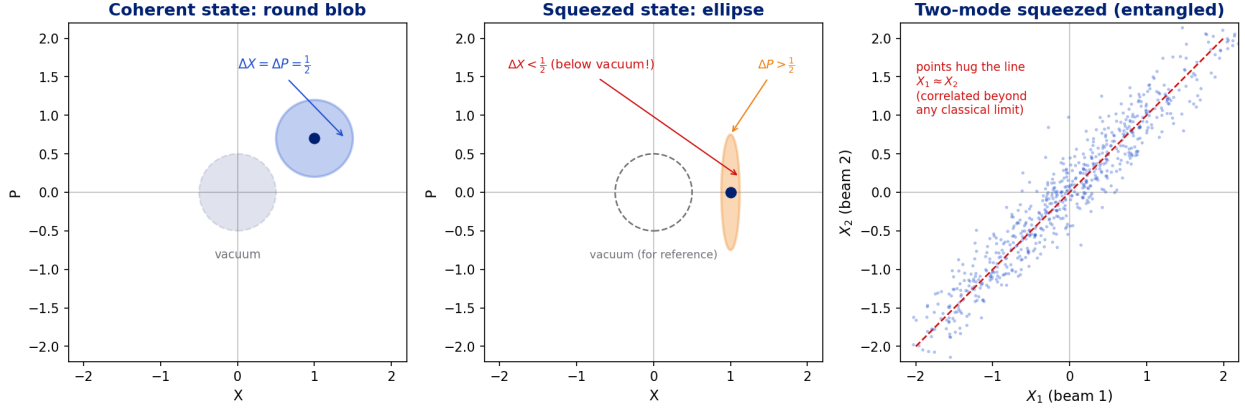


Figure 5: Non-classical light in phase space: a coherent state is a round blob (left); a squeezed state is an ellipse, quieter than the vacuum in one quadrature (middle); a two-mode-squeezed (entangled) pair has $X_1 \approx X_2$ correlated beyond any classical limit (right).

CV entanglement, shown mathematically (one example). The CV Bell pair is the *two-mode-squeezed vacuum* (TMSV / “EPR” state). In the Fock basis, for modes A, B ,

$$|\text{TMSV}\rangle = \sqrt{1 - \lambda^2} \sum_{n=0}^{\infty} \lambda^n |n\rangle_A |n\rangle_B, \quad \lambda = \tanh r,$$

with r the squeezing strength. **Why it is entangled** (same argument as the Bell state above): if it factored as $|\psi\rangle_A \otimes |\phi\rangle_B$ with amplitudes $c_{nm} = a_n b_m$, then $c_{00} \neq 0$ and $c_{11} \neq 0$ force a_0, a_1, b_0, b_1 all nonzero—but then $c_{01} = a_0 b_1 \neq 0$, contradicting $c_{01} = 0$ (the state has only equal- n terms). No product form exists $\Rightarrow$ entangled. Physically this is exactly EPR: in the large- r limit the quadratures become perfectly correlated, $X_A - X_B \rightarrow 0$ and $P_A + P_B \rightarrow 0$, tighter than $\Delta X \Delta P \geq \frac{1}{4}$ allows for two *independent* beams. This is the resource consumed by CV teleportation (§6).

6. Quantum teleportation — moving a state, not matter

Teleportation transfers an *unknown* qubit $|\psi\rangle$ from Alice to Bob *without sending the particle*, by consuming one shared entangled pair plus two classical bits. It is genuinely *physical*—and *not* faster than light, because Bob can do nothing useful until the two classical bits arrive (Fig. 6).

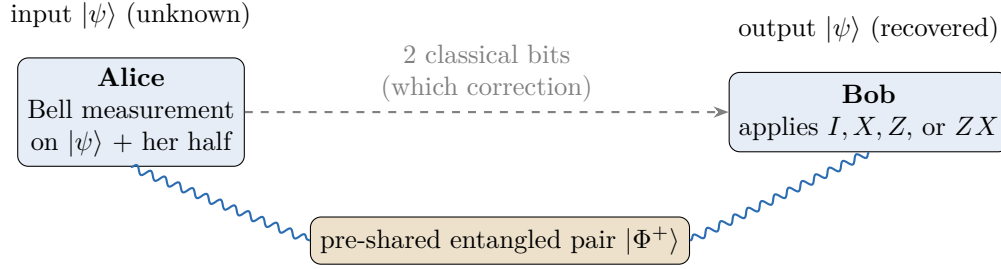


Figure 6: Quantum teleportation. Alice makes a Bell measurement on the unknown $|\psi\rangle$ and her half of a pre-shared pair, sends 2 classical bits, and Bob applies the matching Pauli correction to recover $|\psi\rangle$ — no particle travels from Alice to Bob.

The exact experiment (step by step).

1. **Resource.** Alice and Bob *pre-share* an entangled pair: qubit A with Alice, qubit B with Bob, in state $|\Phi^+\rangle_{AB}$. (These two photons were sent out ahead of time, one to each, as in §5.)
2. **Alice's input.** Alice also holds the unknown qubit $C = |\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ she wants to send. She does *not* know α, β .
3. **Alice's operation.** She performs a *joint Bell measurement* on *her two qubits* (C, A). The outcome is one of four possibilities = **two classical bits**. This measurement *destroys* $|\psi\rangle$ at her end.
4. **What is transmitted.** Only those **2 classical bits** travel to Bob (ordinary channel). No quantum particle is sent from Alice to Bob.
5. **Bob's operation.** Depending on the 2 bits, Bob applies one Pauli correction $\{I, X, Z, ZX\}$ to *his* qubit B , which then becomes exactly $|\psi\rangle$.

The regrouping, worked. Start from $|\psi\rangle_C \otimes |\Phi^+\rangle_{AB}$ and rewrite the two qubits Alice will measure, (C, A) , in the Bell basis $\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}$. Pure algebra (no physics yet) gives the identity

$$|\psi\rangle_C |\Phi^+\rangle_{AB} = \frac{1}{2} \left[|\Phi^+\rangle_{CA} (\alpha|0\rangle + \beta|1\rangle)_B + |\Phi^-\rangle_{CA} (\alpha|0\rangle - \beta|1\rangle)_B \right. \\ \left. + |\Psi^+\rangle_{CA} (\alpha|1\rangle + \beta|0\rangle)_B + |\Psi^-\rangle_{CA} (\alpha|1\rangle - \beta|0\rangle)_B \right].$$

Read off Bob's qubit in each line: it is $|\psi\rangle$ up to a known Pauli $\{I, Z, X, XZ\}$ respectively. So whichever of the four equally likely Bell outcomes Alice gets, she tells Bob (2 bits) which correction to undo, and Bob recovers $|\psi\rangle$ *exactly*. The original C was consumed in the Bell measurement—so only *one* copy ever exists, consistent with no-cloning.

Difference between §5 (entanglement) and §6 (teleportation). §5 only *shares correlated randomness* (a pair, one photon each); by itself it moves no chosen information. §6 *uses* that shared pair, *plus* a Bell measurement at Alice and *2 classical bits* to Bob, to transfer a specific *unknown* state $|\psi\rangle$ without sending the particle that carried it. In short: entanglement is the fuel; teleportation is one engine that burns it.

Teleportation works in CV too. The same recipe runs on continuous variables: Alice and Bob share a two-mode-squeezed (EPR) pair, Alice makes a *joint homodyne* measurement on her mode and the unknown input, sends the two *continuous* outcomes classically, and Bob *displaces* his mode by those amounts to reconstruct the state. This was the first *unconditional* teleportation experiment (Furusawa *et al.*, *Science* 1998, with squeezed light). So teleportation is not DV-only—the correction is a continuous displacement instead of a discrete Pauli.

CV teleportation, shown mathematically (one example). Take the ideal (infinite-squeezing) limit of the shared TMSV pair, so $X_A = X_B$ and $P_A = -P_B$. Alice holds the unknown input mode “in” with quadratures $(X_{\text{in}}, P_{\text{in}})$.

1. **Joint homodyne (Alice’s “Bell measurement”).** Alice measures two commuting combinations, getting two *real numbers* $u = X_{\text{in}} - X_A$ and $v = P_{\text{in}} + P_A$, and sends (u, v) to Bob classically (the CV analog of 2 classical bits).
2. **Displacement (Bob’s Pauli analog).** Bob shifts his mode B by (u, v) :

$$X'_B = X_B + u = X_A + (X_{\text{in}} - X_A) = X_{\text{in}}, \quad P'_B = P_B + v = -P_A + (P_{\text{in}} + P_A) = P_{\text{in}}.$$

Bob’s mode is now *exactly* the input—no photon travelled from Alice to Bob, only (u, v) plus the pre-shared entanglement. Dictionary: Bell pair $\rightarrow$ TMSV; Bell measurement $\rightarrow$ joint homodyne; 2 bits $\rightarrow$ 2 reals; Pauli correction $\rightarrow$ displacement. (At finite r , Bob’s output carries extra noise $\propto e^{-2r}$ —the reason real fidelities are < 1 .)

7. Decoherence and quantum coding — protecting real photons

Loss vs. decoherence. Two different things happen to real photons in a channel:

- **Loss:** the photon is *absorbed or scattered* and never arrives. In telecom fibre the attenuation is about 0.2 dB/km at 1550 nm, so photons are lost *probabilistically*—some arrive, many do not. (If *all* were lost there would be no link; the surviving fraction carries the communication.)
- **Decoherence:** the photon arrives but its delicate *superposition/phase is scrambled* by coupling to the environment—fibre birefringence drifting the polarisation, temperature and vibration shifting the phase (time-bin), stray background light and detector dark counts adding noise. Decoherence turns a clean qubit into a noisy one; it is the quantum version of a corrupted bit.

What happens end to end (a concrete example). *At Alice:* she prepares a logical qubit and, to protect it, *encodes* it across several physical photons using a quantum error-correcting code (e.g. the Steane $[[7, 1, 3]]$ code: 1 logical qubit spread over 7 physical qubits). *In the channel:* loss removes some photons and decoherence flips/dephases others. *At Bob:* he measures *syndromes*—extra parity checks that reveal *where* an error occurred **without revealing the logical data**—and applies a unitary (measurement-free) correction. The logical qubit is restored. You *cannot* “measure and re-send” to fix a quantum signal: measuring the data would collapse it and would violate no-cloning. Over long links, quantum *repeaters* and *entanglement purification* play the analogous role.

Why Michel’s coding expertise transfers—with a twist. A quantum code *spreads* one logical qubit across several physical qubits and corrects errors *without learning the data*. The *code design* is mathematics (CSS, surface codes—close cousins of classical linear codes). But the *protection acts on real quantum states*, and the operations are *unitary and measurement-free*, not the classical decode-measure-correct. Same intuition, different rulebook.

The quantum channel is not a binary channel—it is a quaternary Pauli channel. A classical bit suffers one kind of error (a flip, $0 \leftrightarrow 1$). A qubit can suffer a *bit flip* (X), a *phase flip* (Z), *both* (Y), or nothing (I)—the four *Pauli* errors. So quantum error correction must fix a richer, quaternary error set, which is why the coding overhead is large and why designing quantum forward-error-correction (QFEC) that approaches the *Hashing bound* (the quantum capacity limit) is a central challenge. As Prof. Hanzo puts it, “coding is the king of quantum” [13]: the field’s scaling promises ride on how well these codes work—directly the lineage of Michel’s classical coding expertise.

Does error correction work for CV too? Yes—“bosonic” codes. *Bosonic* means built from bosons—quanta of *any* quantum harmonic oscillator: a single *mode* has an *infinite-dimensional* ladder of Fock states $|0\rangle, |1\rangle, \dots$, with quadratures X, P . The oscillator can be the *electromagnetic field* (its quanta are *photons*), a mechanical vibration in a crystal or a trapped ion’s motion (*phonons*), or a microwave-cavity field. **Do we need phonons? No.** *Phonons are just one example of a bosonic mode*; the GKP/cat states in this figure and everything Nairi will build are *optical/microwave photons*. We mention phonons only because bosonic codes were first demonstrated in both photon and phonon platforms—in *our* CV work (coherent states, homodyne, telecom fibre) the mode is always the **photonic field**, so phase coherence lives in the light, not in any sound wave. CV quantum information lives in such bosonic modes, and *bosonic codes* protect a logical qubit/qudit *inside* one or more oscillators.

- **How any stabilizer code works (DV and CV alike).** Choose commuting operators, the *stabilizers* S_i ; the code space is their simultaneous $+1$ eigenspace. *Logical* operators commute with every S_i (so they act on the data without disturbing the checks). An error anticommutes with, or shifts, some S_i . You *measure the* S_i —a projective measurement that *collapses onto the syndrome* (the eigenvalues of the stabilizers), revealing the error *but not the logical data*—then undo it.
- **DV example (discrete syndrome).** Three-qubit code, stabilizers $Z_1 Z_2$ and $Z_2 Z_3$ (parity checks). A bit-flip on qubit 2 flips both parities $\rightarrow$ syndrome $(-1, -1) \rightarrow$ flip it back. The syndrome is a *discrete* ± 1 .
- **CV example (continuous / modular syndrome): GKP codes.** Encode a qubit in an oscillator as a grid of spikes in the X – P plane. The stabilizers are *displacements* by $2\sqrt{\pi}$; measuring X and P *modulo* $\sqrt{\pi}$ collapses onto the *fractional shift* (a continuous/modular syndrome), telling you how far off the grid you are—without revealing which logical state (that lives in the grid’s global structure). Displace back to correct.
- **Other bosonic codes.** *Cat codes* (a qubit in superposed coherent states) and *binomial codes* fight the dominant CV error—*photon loss*—by detecting a change in photon-number *parity* (a QND measurement); Yale reached “break-even” with a cat code.
- **The Gaussian no-go (why CV-QEC is hard, not impossible).** The channel’s noise

(loss + thermal + vacuum) is *Gaussian*—imposed by physics, not chosen by us. A theorem (Niset–Fiurášek–Cerf, 2009) proves you *cannot* correct Gaussian errors using only *Gaussian* operations (beam splitters, squeezers, homodyne). So CV-QEC *requires a non-Gaussian resource*: a photon-number-resolving detector, photon subtraction/addition, a Kerr nonlinearity, or a qubit coupled to the cavity (how GKP/cat states are actually made). Hard and lossy—today a frontier, mostly for CV *computing* and repeaters, not the bare link.

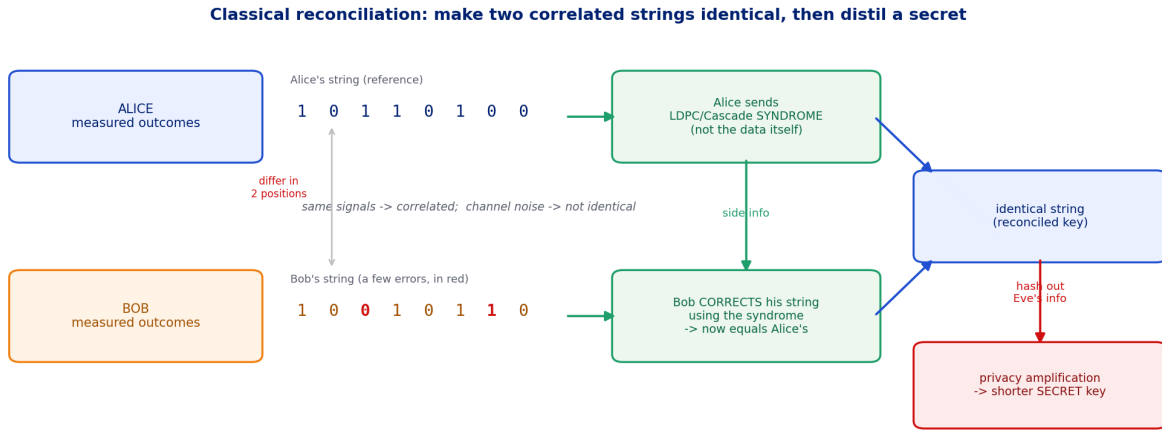
- **Two independent coding layers (don’t confuse them).** (1) QEC *on the quantum states* (qubit stabilizer codes for DV; bosonic codes for CV)—used in memories/repeaters/computing, and *normally absent on a bare QKD/QSDC link*. (2) *Classical reconciliation coding* (LDPC/Cascade), which acts *directly on the measurement outcomes*—not on a decoded QEC codeword. The usual link flow is therefore *measure* $\rightarrow$ *reconcile* $\rightarrow$ *privacy-amplify*, with *no* quantum-QEC step feeding the classical decoder. Both DV and CV use layer (2); the difference is the data type—DV reconciles *bits*, CV reconciles *continuous* values (slice / multidimensional reconciliation, harder). Layer (2) is where classical coding expertise plugs straight in (Fig. 7).

Note: teleportation and entanglement are *not* DV-only—*CV teleportation* (Furusawa 1998, using two-mode-squeezed “EPR” light) and CV entanglement are standard. And a *qudit* is a d -level system ($d > 2$, e.g. qutrit $d = 3$); qudits are still discrete (DV-family), but GKP can encode a qudit into an oscillator, bridging qudits and CV.

How GKP bridges “qudit” and CV (with the math). Native CV is *not* a qudit—it carries *continuous* (X, P) , an infinite-dimensional continuum, not a discrete d -level alphabet. GKP is the bridge: it hides a clean d -level system *inside* one oscillator by making the code space invariant under a *lattice* of phase-space displacements. Write the displacement operators $\hat{D}_X(s) = e^{-2is\hat{P}}$ (shift in X by s) and $\hat{D}_P(s) = e^{2is\hat{X}}$ (shift in P by s).

- **Qubit GKP ($d = 2$).** Stabilizers are the two commuting displacements $S_X = \hat{D}_X(2\sqrt{\pi})$ and $S_P = \hat{D}_P(2\sqrt{\pi})$; the code states are grids of spikes spaced by $2\sqrt{\pi}$. Logical operators are the *half-cell* shifts $\bar{X} = \hat{D}_X(\sqrt{\pi})$, $\bar{Z} = \hat{D}_P(\sqrt{\pi})$ —so $|0\rangle_L$ and $|1\rangle_L$ are two *interleaved* sublattices offset by $\sqrt{\pi}$ (exactly the blue/orange grids in panel (c) of the code figure).
- **Qudit GKP (general d).** Keep the *same* idea but pack d interleaved sublattices into each unit cell: the logical shift becomes $\hat{D}_X(\sqrt{4\pi/d})$, so d distinct grids fit between two stabilizer spikes. Measuring X and P *modulo* the cell spacing gives the (continuous) syndrome—how far you drifted—while *which of the d sublattices* you sit on carries the logical qudit value. One oscillator, d logical levels.

Takeaway: a qudit is an *abstract* d -level system; GKP *realizes* it in a CV mode. So “no native qudit for CV” is right—but GKP embeds one anyway, which is why the same $Q(q, d)$ qudit-coding language can eventually target CV hardware.



Key points:

- Acts directly on the MEASUREMENT OUTCOMES (bits for DV; continuous values for CV). There is NO quantum-QEC step on the link.
- It is side-information coding (Slepian-Wolf): the code's syndrome fixes the discrepancy; then hashing removes whatever Eve could know.

Figure 7: **Classical reconciliation** (information reconciliation), the layer (2) coding above. After measuring, Alice and Bob hold *correlated but not identical* strings—correlated because they came from the same signals, different because of channel noise (here Bob’s copy differs in two positions, in red). Alice sends the *syndrome* of a classical code (LDPC/Cascade)—*not* the data—and Bob uses it as side information to correct *his own* string to match Alice’s (Slepian–Wolf side-information coding). *Privacy amplification* then hashes the reconciled string down to a shorter secret key, removing whatever an eavesdropper could have learned. Crucially this runs *directly on the measurement outcomes*: there is *no* quantum-QEC step on the link feeding it.

Three code examples: same idea (redundancy + syndrome), three carriers -- qubit, qudit, oscillator

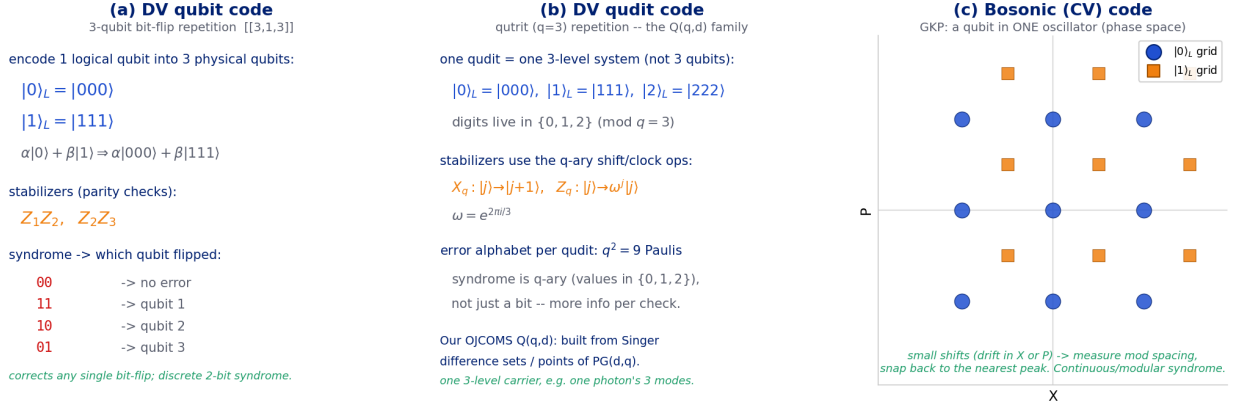


Figure 8: Three worked code examples, one per carrier. **(a)** A *DV qubit* code—the $[[3,1,3]]$ bit-flip repetition code: one logical qubit spread over three physical qubits, protected by two parity stabilizers $Z_1 Z_2, Z_2 Z_3$ whose *discrete 2-bit* syndrome names the flipped qubit. **(b)** A *DV qudit* code—qutrit ($q=3$) repetition: *one* three-level carrier (not three qubits), digits in $\{0, 1, 2\} \pmod{3}$, stabilizers built from the q -ary shift/clock operators X_q, Z_q ; the syndrome is now q -ary (carries more information per check). This is the family our OJCOMS codes $Q(q, d)$ generalize (from Singer difference sets / points of $PG(d, q)$). **(c)** A *bosonic (CV)* code—GKP: a qubit encoded in *one* oscillator as a grid of phase-space peaks ($|0\rangle_L$ and $|1\rangle_L$ interleaved), where small X/P drifts are corrected by measuring position *modulo* the grid spacing and snapping to the nearest peak (a *continuous/modular* syndrome). Same idea throughout—redundancy plus a syndrome—realized on a qubit, a qudit, and an oscillator.

One line to carry away (Fig. 8). The *principle* of quantum error correction is identical for all three: add redundancy, measure stabilizers to get a *syndrome* without disturbing the logical information, and correct. What changes is only the *alphabet of the syndrome*—one bit per check (qubit), one q -ary digit per check (qudit), or a continuous value read *modulo* a lattice spacing (bosonic/GKP). Our coding line with Prof. Hanzo lives in the middle panel (qudit stabilizer codes); a guesswork (GRAND) decoder is the natural next step because the stabilizer check *is* the code-membership test GRAND needs.

8. The electromagnetic spectrum — radio, microwave, light are one family

Much confusion (“is light a radio wave? what is microwave? what is ‘electromagnetic’?”) dissolves with one picture (Fig. 9). **An electromagnetic (EM) wave is an oscillating electric field paired with an oscillating magnetic field, travelling at the speed of light c .** “Electric” and “magnetic” are the two coupled halves of the *same* wave; you never have a travelling EM wave with only one. Radio, microwave, infrared, visible light, ultraviolet, X-rays and gamma rays are *all the same phenomenon*, differing only in **wavelength** λ (and therefore frequency $f = c/\lambda$ and photon energy $E = hf$, with h Planck’s constant).

When are electric and magnetic fields separate? Only when they are *static* (non-radiating): a charged object or a capacitor has a pure *electric* field; a *permanent magnet*, a DC electromagnet, an MRI coil, a motor or a compass has a pure *magnetic* field. The moment charges *accelerate*, the two lock together and radiate as an EM wave (light/radio), whose quanta are *photons*. So static fields can be purely electric *or* purely magnetic; the propagating EM wave we communicate with is always both, and its energy comes in photons (not electrons: electrons are quanta of the matter field, not of the EM field).

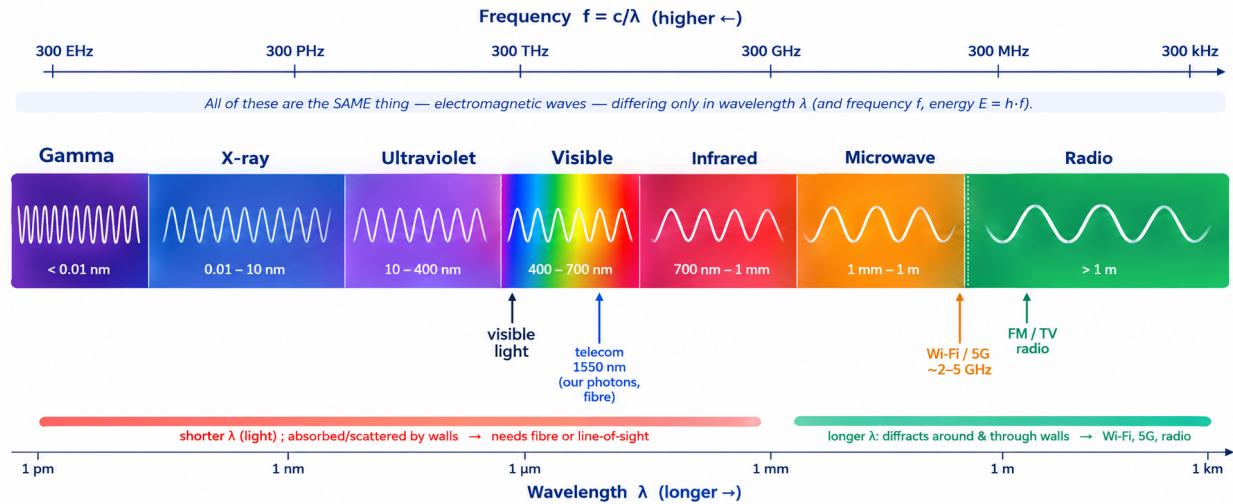


Figure 9: The electromagnetic spectrum. Radio, microwave, infrared, visible, UV, X-rays and gamma rays are the same phenomenon, differing only in wavelength λ (hence frequency $f = c/\lambda$ and photon energy $E = hf$). Longer λ penetrates walls; light needs fibre or line-of-sight.

The three quantities are one. $c = \lambda f$ (constant speed), so long wavelength = low frequency; and $E = hf$, so higher frequency = higher energy per photon. A 1550 nm telecom photon ($f \approx 193 \text{ THz}$) carries about 0.8 eV; a 2.4 GHz Wi-Fi photon carries $\sim 10^5$ times *less* energy. That energy gap is why single *optical* photons are easy to detect one-by-one (DV), while single *microwave* photons are so faint they are swamped by room-temperature heat and need cryogenics.

Why wavelength decides what penetrates walls (the key practical fact). Long-wavelength waves (radio, microwave: metres to centimetres) *diffract around* obstacles and pass *through* walls—this is why Wi-Fi, 5G and radio work indoors and around corners. Short-wavelength waves (visible and near-infrared *light*: sub-micron) are *absorbed and scattered* by ordinary materials, so light does *not* go through walls; it must travel in a transparent *fibre* or along an unobstructed *line of sight*. Quantum communication rides on light (§9), which is precisely why “you cannot send single photons to a moving phone through walls” (§13).

9. Physical layer I: the carrier and the channel

Separate *what the qubit rides on* (carrier) from *the medium it travels through* (channel).

9.1 Carrier: which photons

- **Optical vs. telecom light.** “Optical” means light in general (visible 400–700 nm plus near-infrared). “Telecom light” means the specific near-infrared windows where silica fibre is clearest: the *O-band* near 1310 nm (lowest dispersion) and the *C-band* near 1550 nm (lowest loss, ~ 0.2 dB/km). Visible light has *high* loss in fibre, so it is used for free-space and lab work, not long fibre. Parameter snapshot: 1550 nm $\leftrightarrow$ 193 THz $\leftrightarrow$ 0.2 dB/km; 500 nm (green) $\leftrightarrow$ 600 THz $\leftrightarrow$ many dB/km in fibre.
- **Microwave photons.** These exist ($\sim$ GHz, wavelength cm–mm) and are the native qubits of superconducting quantum *computers*. But their photon energy is $\sim 10^5\times$ smaller than optical, so at room temperature thermal photons ($kT \gg hf$) drown them. They therefore work only in *cryogenic* lines. *Cryogenic* means cooled to near absolute zero (a dilution refrigerator, ~ 0.01 K), which removes the thermal photons. Even then, loss in a cooled cable is huge over distance, so microwave-photon links are *chip-to-chip* (centimetres to metres inside/between cryostats)—great for wiring quantum processors, useless for long-range or wireless. Yes, it is simply a much longer wavelength than optical; both are EM waves (§8).

9.2 Channel: only two viable media, and why

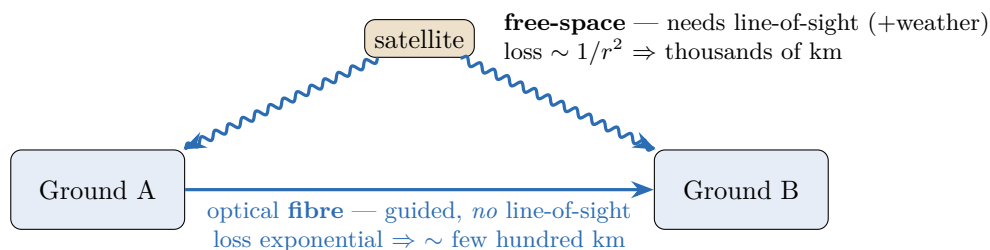


Figure 10: The two viable quantum channels: guided optical fibre (no line-of-sight, but exponential loss limiting reach to a few hundred km) and free-space line-of-sight including satellite ($1/r^2$ loss, reaching thousands of km).

The carrier is light, and (from §8) light does not go through walls or around the horizon. So there are exactly two options (Fig. 10): a **guided** medium—*optical fibre*, which needs *no* line-of-sight but whose loss grows exponentially with length—or **free-space line-of-sight** (ground-to-ground or ground-to-satellite), which reaches thousands of km but needs a clear path and good weather. There is no “quantum Wi-Fi” through walls: a photon that hits a wall is absorbed (loss) or scrambled (decoherence), destroying the qubit. China’s **Micius** satellite distributed entangled photons over 1,200 km and a key over 7,600 km by trusted relay [7, 8]. Both fibre and free-space can carry DV or CV signals.

Two active research frontiers (where our SDR work points). (i) *Simultaneous classical and quantum communication*: carrying classical data *and* a quantum signal over one set of transceivers / the same waveform, rather than separate links [14]. (ii) *Pushing quantum beyond optical into RF*: while long-range quantum today is optical, there is active work on THz, mmWave and even microwave-band quantum links [14, 15], and on folding quantum into cellular infrastructure (“1Q”: quantum base stations distributing entanglement alongside radio). This is exactly the seam our I/Q $\leftrightarrow$ X/P bridge and USRP testbed sit on—classical and quantum on the same coherent hardware.

10. Physical layer II: how a qubit is written on a photon (encoding)

The abstract $|0\rangle, |1\rangle$ must be mapped onto a real, controllable property (a “degree of freedom”) of the photon [10]. The three most common are shown in Fig. 11:

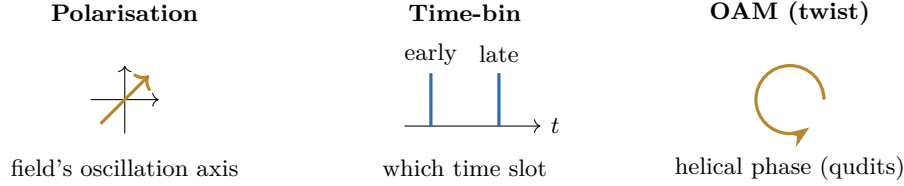


Figure 11: Three common single-photon encodings: polarisation (the field’s oscillation axis), time-bin (which time slot the photon occupies), and orbital angular momentum (helical-phase twist, which enables qudits).

10.1 Polarisation — and how a photon can be “both at once”

Light is a transverse EM wave: its electric field oscillates in a direction *perpendicular* to travel. **Polarisation is that oscillation axis** (it is *not* a magnetic north/south pole). Map horizontal = $|0\rangle$ and vertical = $|1\rangle$. A diagonal photon is the genuine *superposition* $|+\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$ —its field oscillates along 45° , which is literally horizontal *and* vertical at once, not “sometimes H, sometimes V.”

Answering: “once polarised, doesn’t it stay fixed, like magnetisation?” A photon’s polarisation state *is* stable in flight (that is why it can carry a bit)—but it can be *prepared in a superposition* and it stays a superposition until measured. Unlike a magnetised needle that has settled to one direction, a diagonal photon has *no* single H-or-V value; it is in both. If Bob measures it in the H/V basis he gets H or V with probability $\frac{1}{2}$ and the state *collapses*; if he measures in the diagonal basis he gets “diagonal” with certainty. Superposition is a real, preparable condition, not our ignorance. Fibre scrambles polarisation, so long fibre links prefer time-bin/phase instead.

10.2 Time-bin — superposition in time

Send the photon through an *unbalanced interferometer* (a short arm and a long arm). It can leave in an *early* slot ($|0\rangle$) or a *late* slot ($|1\rangle$)—or, if it takes “both arms,” in the superposition $\alpha|early\rangle + \beta|late\rangle$: the photon genuinely has amplitude in *both* time slots. Bob passes it through a *matching* interferometer so early-long and late-short overlap and *interfere*; his measurement collapses the result to early, late, or an interference outcome. Time-bin is robust in fibre (immune to polarisation drift), which is why fibre QSDC/QKD favours it.

10.3 OAM / helical phase — twist, and qudits

A normal beam has a flat wavefront. An **orbital-angular-momentum** beam has a wavefront that *spirals like a corkscrew*: its phase winds by $2\pi\ell$ around the axis, described by $e^{i\ell\varphi}$, where the integer *topological charge* ℓ is the number of twists. That ℓ is the physical carrier: $\ell = 0, \pm 1, \pm 2, \dots$ gives *many* distinguishable states, so one photon can carry *more than one bit* (a “qudit”). It is created by passing light through a *spiral phase plate*, a fork hologram, or a spatial light modulator;

a single photon can be placed in a *superposition* of twists just as it can of polarisations. OAM is powerful but fragile (atmospheric turbulence distorts the twist), so it is a free-space, not fibre, technique.

Are these the only ways? No—but they are the natural photonic “knobs.” Besides polarisation, time-bin and OAM, one can encode in *frequency/colour* (frequency-bins) or in *path* (which of two fibres, “dual-rail”). A photon only has a handful of controllable degrees of freedom, and these are them; the choice is dictated by the channel (fibre vs. free-space) and the hardware. Encodings can even be combined (hybrid / hyper-entanglement).

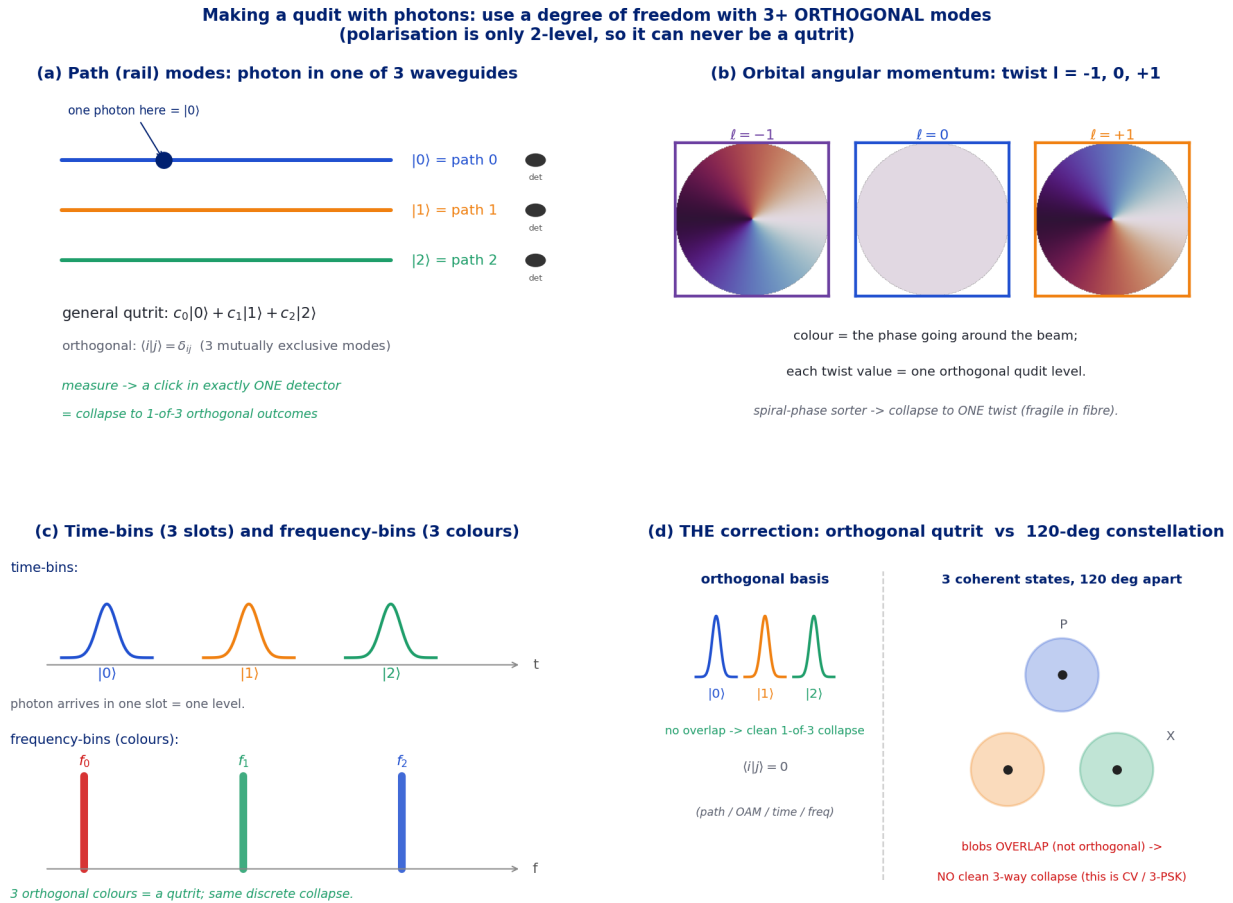


Figure 12: Making a DV qudit ($d = 3$) with photons—and a common misconception corrected. (a) *Path/rail* modes: a photon in one of three waveguides realizes $|0\rangle, |1\rangle, |2\rangle$; a general qutrit is $c_0|0\rangle + c_1|1\rangle + c_2|2\rangle$ with *orthogonal* basis states ($\langle i|j\rangle = \delta_{ij}$), and measurement gives a click in *exactly one* detector—collapse to 1-of-3. (b) *OAM*: the twist $\ell = -1, 0, +1$ (colour = phase winding around the beam) gives three orthogonal levels. (c) *Time-bins* (three slots) and *frequency-bins* (three colours) do the same discretely. (d) *The correction*: orthogonal qutrit states (left) do *not* overlap, so measurement collapses cleanly to one of three; a set of *three coherent states* 120° *apart in the X–P plane* (right) is a different object—a CV/3-PSK *constellation* whose blobs *overlap* (non-orthogonal), so it does *not* give a clean three-way collapse. A qudit needs a degree of freedom with ≥ 3 *orthogonal* modes; *polarisation is only 2-level* (the Poincaré sphere is a qubit) and can never be a qutrit.

Reading Fig. 12: does polarisation collapse to two, and a qutrit to three? Yes for the qubit—a diagonal photon $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$ measured in the H/V basis collapses to H or V , 50/50. But the two outcomes are two *orthogonal polarisations* (H vs V)—*not* “electric vs magnetic”: every photon always carries both an $\mathbf{E}$ and a $\mathbf{B}$ field, and polarisation is just the *direction* $\mathbf{E}$ oscillates. For a qutrit you *cannot* reuse polarisation (it is intrinsically 2-level); you switch to path, OAM, time-bin or frequency-bin, where three *orthogonal* modes give a clean 1-of-3 collapse. And “three coherent states 120° apart” is *not* a qutrit—that is a CV constellation (overlapping, non-orthogonal), a different animal from an orthogonal qudit basis.

11. Physical layer III: sources and the two receivers (DV vs. CV)

Sources. Either an *attenuated laser* (weak coherent pulses, made safe with decoy states), or a *true single-photon source*: a **quantum dot** (a nanoscale semiconductor “artificial atom” that emits one photon at a time), an **NV centre** (a nitrogen-vacancy defect in diamond—a trapped electronic spin that emits single photons and can store a qubit), or **SPDC pairs** (§5: use one photon of the pair as a “herald” announcing its twin).

Receivers come in two fundamentally different families (Fig. 13):

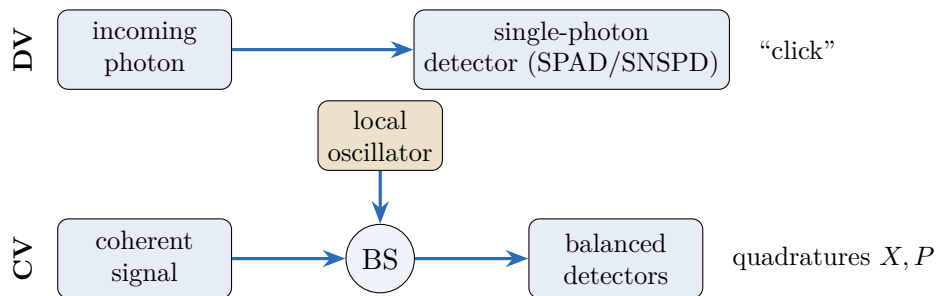


Figure 13: The two receiver families. DV: a single-photon detector (SPAD/SNSPD) that registers “clicks.” CV: a coherent signal beaten against a local oscillator on a beamsplitter (BS) and read by balanced detectors, yielding the quadratures X, P (the same I/Q measurement as an SDR).

DV vs. CV — what actually differs (the question everyone asks). *DV (discrete-variable).* Information rides on a property of a *single* photon (polarisation, time-bin, OAM). You still send *many* photons—one per time slot, millions per second—but each carries *one* qubit, and the receiver is a *single-photon detector* that registers individual “clicks,” photon by photon. It counts particles. *CV (continuous-variable).* Information rides on the *amplitude and phase* of a brighter *coherent* pulse (a little burst of laser light). The receiver is a *homodyne* detector: it mixes the signal with a strong reference laser (the *local oscillator*) on a beamsplitter and reads the field *quadratures* X, P —effectively averaging over *many* photons per symbol. It measures a wave.

“Why X, P for CV but not for a single photon?” A lone photon is a *number (Fock) state* $|1\rangle$: it has a definite *energy* but a completely *random phase*, so its quadrature averages vanish, $\langle X \rangle = \langle P \rangle = 0$, with large, uninformative spread. You *cannot* read a useful amplitude-phase point off one photon. A *coherent state* (laser-like, many photons) does have a well-defined (X, P) —the “blob” of §12. That is why X, P are the language of *bright coherent light* (CV), while single photons are handled as click/no-click (DV). Formally a single photon *has* quadrature operators, but its state carries no useful mean amplitude or phase.

So what is “coherence”? It is a *stable, well-defined phase* of the wave, in time and across the beam—the property that lets amplitude *and* phase be measured (a laser has it; a light bulb does not). It is *not* the spin of an electron or photon. Losing this phase definiteness is *decoherence*. Polarisation-DV does not lean on phase coherence; phase/time-bin and all of CV do, which is why they need a phase reference (the interferometer or the local oscillator).

11.1 Polarisation vs. phase vs. coherence — a 3D picture

These three words are constantly confused, so here they are side by side on the actual electromagnetic wave (Fig. 14). Recall (§8) that light is an EM wave: an electric field $\mathbf{E}$ and a magnetic field $\mathbf{B}$ oscillating *perpendicular* to each other and to the direction of travel.

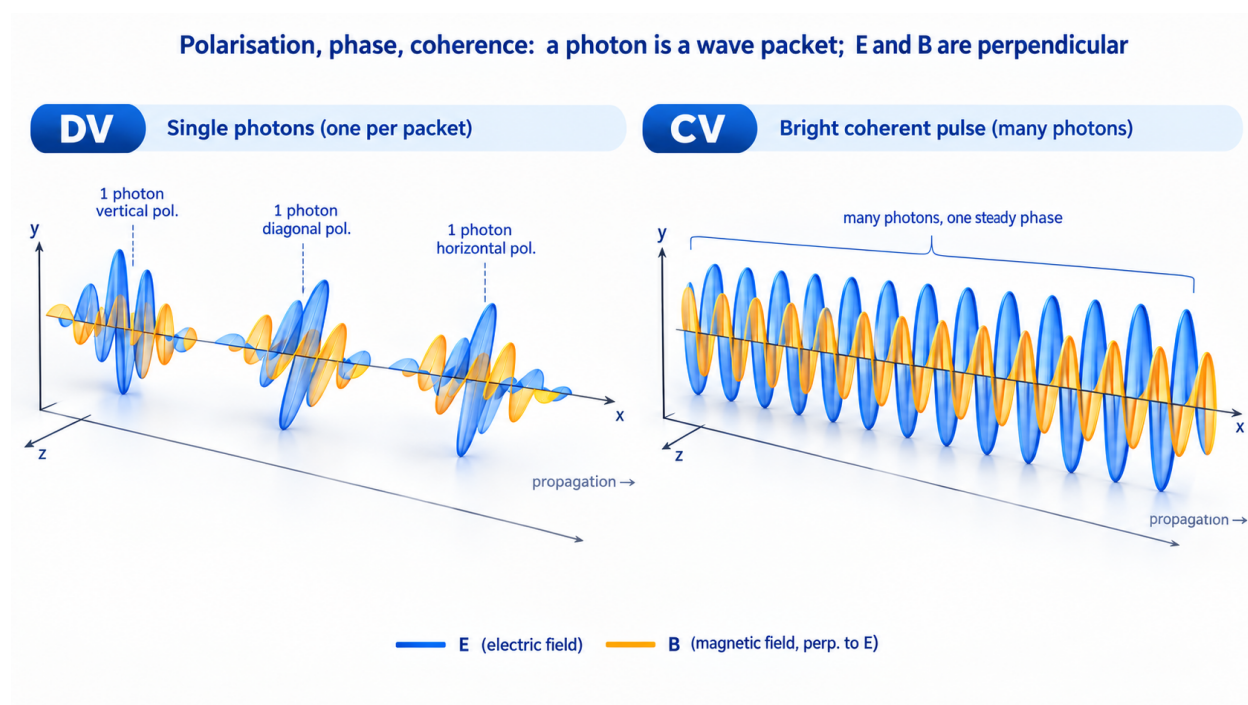


Figure 14: Polarisation, phase and coherence on the actual EM wave (electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ perpendicular). *DV* (left): each symbol is one faint photon — a wave packet — whose polarisation (oscillation axis) carries the bit. *CV* (right): each symbol is a bright coherent pulse of many photons sharing *one steady phase*, read as continuous quadratures X, P .

Direct answers to the exact confusions.

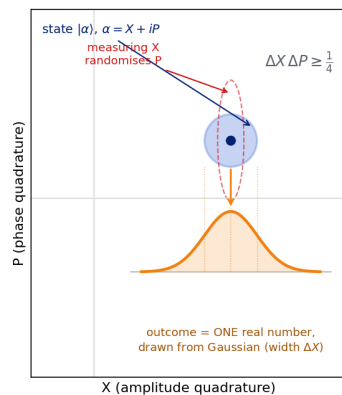
1. **Three different things.** *Polarisation* = the *direction* $\mathbf{E}$ oscillates (the tilt of the wave's plane). *Phase* = the *timing* of the oscillation (where the crest sits along the axis). *Coherence* = the phase being *stable and well-defined* over time and across the beam. A laser is coherent; a light bulb is not. **Polarisation staying fixed is *not* coherence**—coherence is about the phase.
2. **Does each photon have a phase?** A single photon has a *polarisation*, but *not* a well-defined absolute *phase*: its phase relative to any reference is random. A well-defined phase is a property of a *bright coherent pulse* (many photons sharing one mode). *This* is the deep reason CV uses coherent pulses and DV uses single photons.
3. **So what really differs, DV vs. CV? Not polarisation** (both can use any polarisation). It is: *DV* = *one photon per symbol*, information in a discrete property (polarisation/time-bin), detector counts individual *clicks*, and (for polarisation) *no phase coherence is needed*; *CV* = *a bright coherent pulse per symbol*, information in the continuous amplitude & phase (X, P), read by homodyne against a phase reference, and *coherence is essential*.
4. **Must successive photons share polarisation/phase?** In *DV*, **no**—successive photons carry *different* bits, so they deliberately differ in polarisation, and their phases are unrelated (left panel: three packets, three tilts, random phases). In *CV*, successive pulses *share* a phase reference (the local oscillator), and amplitude & phase *are* the data (right panel: one steady phase).
5. **“Same phase but different polarisation—does that destroy the superposition?”** This mixes two separate ideas. *Superposition* lives *inside one qubit* (a diagonal photon is $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$ —H and V at once). *Coherence* is a relationship of the wave's phase *across* time/segments. Two successive photons having different polarisations are simply two different symbols; that does *not* destroy any superposition.
6. **Is a photon a slice of the wave?** Closer: a photon is a *wave packet* (a localised pulse of the field). In DV one symbol is $\approx$ one faint packet (≈ 1 photon); in CV one symbol is a brighter pulse containing *many* photons that share one phase.

	Discrete-Variable (DV)	Continuous-Variable (CV)
Measures	individual photons (particle, “clicks”)	amplitude & phase quadratures X, P (wave)
Hardware	single-photon detectors (SPAD, SNSPD)	homodyne / heterodyne (I/Q) detectors
Cost / temp.	expensive, often cryogenic	telecom parts, room temperature
Reach / rate	longer reach, mature security proofs	very high rate at metro range
Bridge to us	—	<i>same I/Q maths as SDR</i> (§12)

Can Alice and Bob choose either? Yes. DV and CV are *engineering choices* of how to *encode and measure*, not two different natures of light—the underlying quantum field is the same. A given link commits to one (or a hybrid). *DV advantages:* longest reach, works at ultra-low light, strongest security proofs. *CV advantages:* room-temperature telecom hardware, very high key rate at metro distances, and—crucially for us—it is *I/Q detection*. Beyond DV/CV there are protocol *families* (MDI-QKD, TF-QKD, DPS/COW; §2) that layer on top of either.

CV is still quantum: measurement forces collapse -- but onto a CONTINUOUS eigenvalue, not a discrete bit

(a) CV collapse: homodyne reads $X \rightarrow$ one continuous value



(b) Discrete (DV) vs continuous (CV) collapse

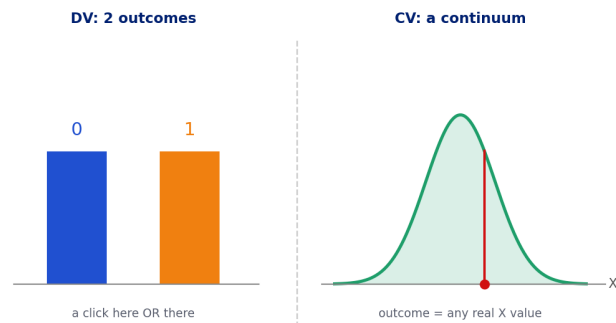


Figure 15: **How a CV state “collapses.”** (a) The state $|\alpha\rangle$ ($\alpha = X + iP$) is a fuzzy blob with $\Delta X \Delta P \geq \frac{1}{4}$. A homodyne measurement of X *projects* the blob onto the X -axis: the outcome is *one continuous real number* drawn from a Gaussian of width ΔX centred on the mean, and the conjugate P is randomised (you measure X or P , never both sharply). (b) So DV collapses to a *discrete* outcome (0 or 1), while CV collapses to a value on a *continuum*—infinitely many possible outcomes. Note the bound is $\geq \frac{1}{4}$ (a floor you cannot beat), not $< \frac{1}{4}$: the blob’s area cannot shrink below the vacuum limit.

Is CV still “quantum”? Yes—collapse, in continuous form (Fig. 15). (A frequent and fair question.)

- **Collapse is general.** Measuring an observable makes the state jump to an eigenstate of *that* observable. *DV*: a *discrete* observable (polarisation) gives a discrete outcome. *CV*: a *continuous* observable (a quadrature X , via homodyne) collapses the state to a *definite* X *value* (a continuous eigenvalue), while the conjugate P is randomised ($\Delta X \Delta P \geq \frac{1}{4}$). Measurement is a *projection* (non-unitary), not unitary evolution. So CV has collapse, disturbance and no-cloning, exactly like DV—it is fully quantum.
- **A coherent pulse is one wave with an uncertain photon number.** $|\alpha\rangle$ is a *single mode* of the field whose photon number is *Poisson-distributed* (mean $|\alpha|^2$). Amplitude and phase (X, P) belong to the *mode*, not to individual photons. The “many photons” are indistinguishable quanta of that *one* wave; they are *not* separate little emitters sitting together, each radiating its own E and B. There is one mode = one E/B wave; “photon number” is just how much energy is in it.
- **Number–phase uncertainty** (heuristically $\Delta N \Delta \phi \gtrsim \frac{1}{2}$). A definite photon number (a

single photon / Fock state) forces a random phase; a well-defined phase (a coherent pulse) forces an uncertain number. That is exactly why DV encodes in a discrete property of *one* photon, and CV encodes in the amplitude/phase of a *coherent pulse*.

- **An eavesdropper’s intercept-resend** therefore perturbs the *mode’s* X, P (adds excess noise to the whole pulse), not “each photon.”
- **Why photons, not electrons?** Communication needs a *flying* qubit: fast (c), low-loss, weakly interacting (low decoherence), telecom-mature, and bosonic (coherent states / lasers exist). Electrons are charged, scatter strongly, decohere fast, and do not travel far freely—excellent as *stationary* qubits (memory, processing) but poor as carriers. Hence *light* for communication, *matter* for memory/computing.

How many bits does a symbol carry? *DV*: one qubit rides one photon, so 10 qubits = 10 photon time-slots (in practice more, because loss drops many). *CV*: bits per symbol are set *exactly as in classical digital comms*—by the modulation and SNR, *not* by counting photons. Discrete modulation: you pick the constellation (QPSK = 2, 16-QAM = 4, 256-QAM = 8 bits/symbol). Gaussian modulation: the rate is the mutual information, $\approx \frac{1}{2} \log_2(1 + \text{SNR})$ per quadrature, tuned via the modulation variance V_A . The pulse brightness (mean photon number $|\alpha|^2$) only sets the *power*, hence SNR, hence the achievable bits—an *indirect* link, never a photon count. And a CV symbol is *one continuous-variable state carrying several classical bits, not a bundle of several qubits*.

Qubit vs. qudit vs. CV — clearing this up. A **qubit** has 2 levels; a **qudit** has d levels ($d > 2$: a *qutrit* is $d = 3$). Note “3 qubits” (three 2-level systems) is *not* a qutrit—a qutrit is *one* 3-level system. **A single photon can carry a qudit**: use a degree of freedom with d settings—e.g. OAM twist $\ell \in \{-1, 0, +1\}$ (a qutrit), three time-bins, or three frequency-bins. So a qudit is *still one photon* (or one discrete system), just with more than two labels. **CV is different**: it is not d discrete levels but a *continuous* amplitude/phase (X, P) carried by a *bright pulse of many (uncertain) photons*. In short: *qubit/qudit = a few discrete levels, typically one photon; CV = a continuous value, a bright pulse*. (Which degree of freedom matters depends on the encoding: polarisation-DV ignores phase; time-bin/phase-DV and all CV use phase.)

12. The bridge: classical I/Q $\leftrightarrow$ quantum X/P

This is the point of the whole primer for our team. A classical software-defined radio and a CV quantum receiver describe a signal with the *same two numbers*—amplitude and phase—so mastering I/Q constellations means already speaking the language of continuous-variable quantum communication (Fig. 16).

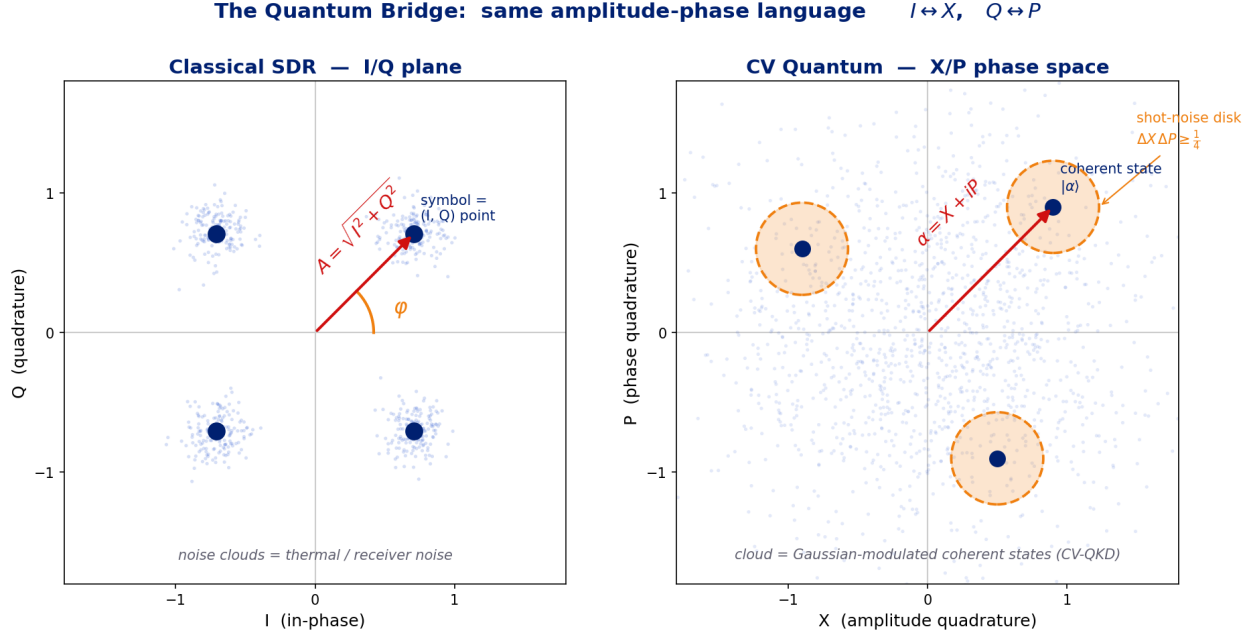


Figure 16: The bridge from classical I/Q to quantum X/P: a software-defined radio and a CV quantum receiver describe a signal with the same two numbers (amplitude and phase), so an I/Q constellation *is* the X, P phase space, with an added irreducible shot-noise floor.

Any wave can be written $E(t) = X \cos \omega t + P \sin \omega t = A \cos(\omega t - \varphi)$ with $A = \sqrt{X^2 + P^2}$ and $\varphi = \arctan(P/X)$. In a radio these two components are called I and Q ; in quantum optics they are the *quadratures* X (amplitude) and P (phase). A homodyne detector measures them exactly as an I/Q mixer does—mix the signal with a reference (local oscillator) at phase 0 to read X , at 90° to read P .

“Is CV really quantum, or just QPSK over a channel?” (The most important question here.) The I/Q *maths* is identical to classical QPSK/QAM—that is the bridge, and why our SDR skills transfer. But CV *is* genuinely quantum; the quantum-ness is *not* in the constellation, it is in four things classical QPSK can never have:

1. **An irreducible vacuum / shot-noise floor** ($\Delta X \Delta P \geq \frac{1}{4}$) that *no* receiver can remove—unlike classical AWGN, which is technical/thermal noise you can cool or average away.
2. **You cannot measure X and P both perfectly:** heterodyne *must* add a full unit of vacuum noise; homodyne destroys the conjugate. Classically you read I and Q with no fundamental penalty.
3. **No-cloning:** a classical QPSK symbol can be copied (amplify/split); a coherent state cannot without adding noise—this is what makes eavesdropping detectable.
4. **Non-classical states:** you can send *squeezed* ($\Delta X < \frac{1}{2}$) or *entangled* CV states; impossible for a classical field.

Operational test: at *high* power (many photons) shot noise is negligible and CV degenerates to ordinary QPSK—effectively “not quantum.” At *low* power (a few photons/symbol) the shot-noise floor + no-cloning + Heisenberg dominate and give provable security. *The quantum-ness*

lives in the low-photon, shot-noise-limited regime, not in the I/Q equations. So CV-over-a-quantum-channel vs. QPSK-over-AWGN share the same equations, but only the former has a hard vacuum floor, no-cloning, and squeezed/entangled states.

Honest consequence for our SDR testbed: an SDR runs at high power with classical, *reducible* noise, so it does *not* have shot-noise-limited, no-cloning security—on SDR, CV is effectively classical QPSK. We therefore *emulate* the protocol/structure; the quantum *guarantee* needs real optical shot-noise-limited hardware. That gap is exactly the *classical–quantum boundary* our eavesdropping-detection study measures.

Shot noise vs. classical AWGN, and non-classical light. *Is shot noise just AWGN?* Statistically they look the same—both roughly white, Gaussian and additive; you cannot tell them apart by shape, and shot noise is usually *modelled* as AWGN. The difference is *physical*: classical AWGN is thermal/electronic noise you can in principle *reduce* (cool the amplifier, average longer); *shot noise is the vacuum fluctuation itself*, an *irreducible* floor set by $\Delta X \Delta P \geq \frac{1}{4}$ that no receiver can beat. It is *not* a matter of coherence—coherence is about the signal’s phase, not the noise. That irreducible floor, together with no-cloning, is what turns “noise” into a *security guarantee*.

Why can’t you measure X and P both perfectly? They are conjugate (non-commuting) observables, like position and momentum. Homodyne at phase 0 reads X exactly but erases P ; to read *both* (heterodyne) you must split the beam, and the beamsplitter’s empty port injects one unit of vacuum noise—so a simultaneous X, P reading always costs extra noise. One mode carries only “one question’s worth” of certainty.

Non-classical light (squeezed, entangled). A coherent state’s noise blob is a *circle* ($\Delta X = \Delta P = \frac{1}{2}$). A *squeezed* state squishes it into an *ellipse*: quieter than the vacuum in one quadrature ($\Delta X < \frac{1}{2}$) at the cost of the other—*below* the classical floor, which no classical light can do (gravitational-wave detectors like LIGO use exactly this). An *entangled* (two-mode-squeezed) pair is two beams whose X, P are correlated beyond any classical limit—the CV cousin of a Bell pair. These exist *only* in the quantum regime; a classical field cannot produce them.

12.1 The coherent-state “blob”

Five terms, defined once.

- **State.** The full quantum description of the system—for one CV mode, the *whole blob* in the X – P plane (its centre *and* its fuzz), not just a pair of numbers.
- **Quantum harmonic oscillator.** The physics of anything that oscillates around equilibrium (a mass on a spring, a pendulum, *a single mode of the light field*). Quantised, its energy comes in equal steps hf ; those steps are the photons. This is the mathematical home of X, P .
- **Fock state $|n\rangle$.** A state with *exactly* n photons (a “number state”): $|0\rangle$ vacuum, $|1\rangle$ one photon, etc. It has a definite energy but a *random phase*.
- **Conjugate (non-commuting) observables.** Two quantities you cannot both pin down at once, because measuring one disturbs the other—formally their operators don’t commute ($XP \neq PX$). X and P are conjugate, exactly like position and momentum ($\Delta x \Delta p \geq \hbar/2$).
- ΔX . The *spread* (standard deviation) of X over many measurements—the blob’s half-width in the X direction, *not* a small change. Large ΔX = fat blob = noisy.

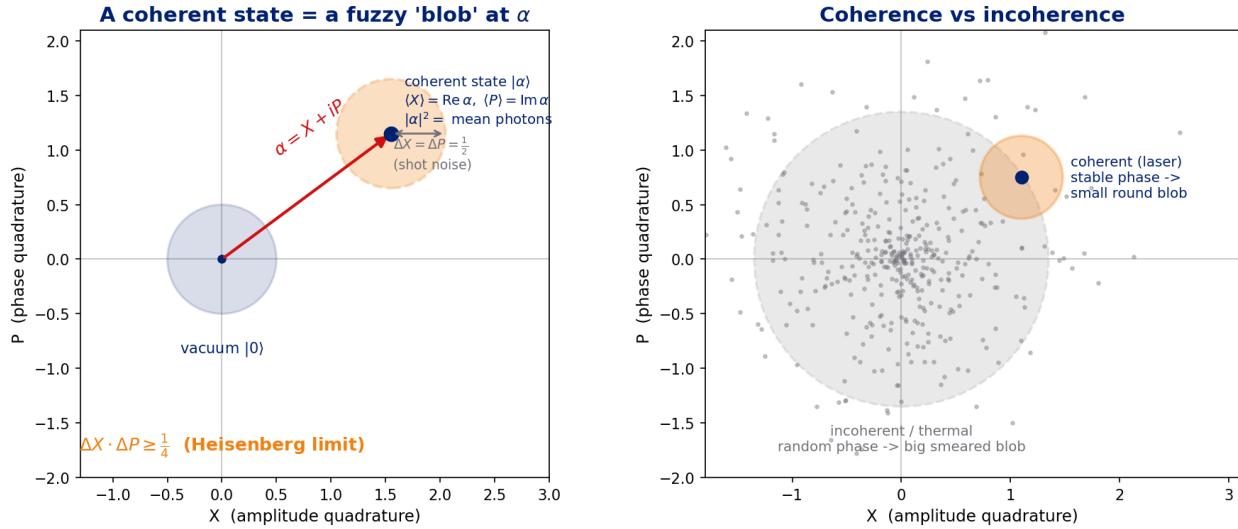


Figure 17: The coherent-state “blob”: a laser pulse sits at a point (X, P) in phase space surrounded by an irreducible fuzziness (shot noise) of area $\Delta X \Delta P \geq \frac{1}{4}$ — the quantum floor beneath every I/Q constellation.

A **coherent state** $|\alpha\rangle$ is the quantum state of an ideal laser pulse, labelled by one complex number $\alpha = X + iP$. Its *means* are $\langle X \rangle = \text{Re } \alpha$, $\langle P \rangle = \text{Im } \alpha$, and $|\alpha|^2$ = the average photon number. So α is the *centre* of a fuzzy dot (a “blob”) in the X – P plane—the value the transmitter *sets* (Fig. 17). A single measurement is a random dot drawn from that blob; the fuzz is the **shot noise**. A *coherent* (laser) state is a small, round, stable blob; an *incoherent* (thermal) state is a big smeared blob with random phase.

Where does $\Delta X \cdot \Delta P \geq \frac{1}{4}$ come from, and where is Planck’s constant? A single mode of the field is mathematically a *quantum harmonic oscillator* with annihilation and creation operators $a, a^\dagger$ obeying $[a, a^\dagger] = 1$. Define the quadratures $X = \frac{1}{2}(a + a^\dagger)$ and $P = \frac{1}{2i}(a - a^\dagger)$. Then a one-line commutator gives

$$[X, P] = \frac{1}{4i}[a + a^\dagger, a - a^\dagger] = \frac{1}{4i}(-2) = \frac{i}{2}.$$

The general (Robertson) uncertainty relation says that for any two observables, $\Delta A \Delta B \geq \frac{1}{2}|\langle[A, B]\rangle|$. Plugging in $[X, P] = i/2$:

$$\Delta X \cdot \Delta P \geq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

The vacuum and every coherent state sit *exactly* at this minimum, with $\Delta X = \Delta P = \frac{1}{2}$ (the shot-noise level). *Where is $\hbar$?* It is still there—the oscillator’s physical position/momentum x, p obey $[x, p] = i\hbar$ and $\Delta x \Delta p \geq \hbar/2$. The quadratures X, P are those variables *rescaled by the zero-point motion* so they are dimensionless; that rescaling *absorbs $\hbar$* , leaving the clean number $\frac{1}{4}$. (Convention warning: some books define $X = a + a^\dagger$, giving $[X, P] = 2i$ and $\Delta X \Delta P \geq 1$ in “shot-noise units.” Same physics, different normalisation—do not be surprised to see “ ≥ 1 ” elsewhere.)

“You cannot make amplitude and phase both perfectly sharp”—the maths. Because X and P do not commute ($[X, P] = i/2 \neq 0$), no state can have both $\Delta X \rightarrow 0$ and $\Delta P \rightarrow 0$: their product is floored at $\frac{1}{4}$. Squeeze one and the other must grow ($\Delta X \Delta P \geq \frac{1}{4}$ still holds—“squeezed light” does exactly this). A coherent state is the balanced minimum, $\Delta X = \Delta P = \frac{1}{2}$. This is a *fundamental* limit (like position/momentum), not a matter of coherence; but coherent states are special precisely because they sit *on* the limit, while thermal/incoherent light sits far above it.

12.2 Shot noise, CV-QKD, and security

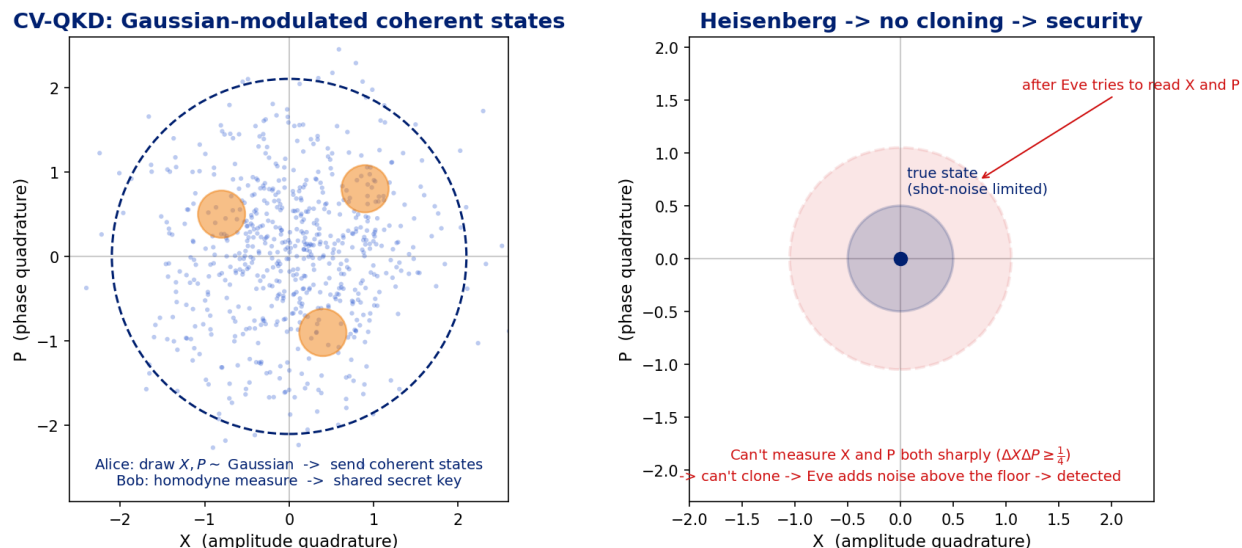


Figure 18: Why the blob gives security: an eavesdropper who measures the coherent state cannot beat the shot-noise limit or clone it, so any intercept-and-resend adds excess noise that Alice and Bob detect — the physical basis of CV-QKD/QSDC security.

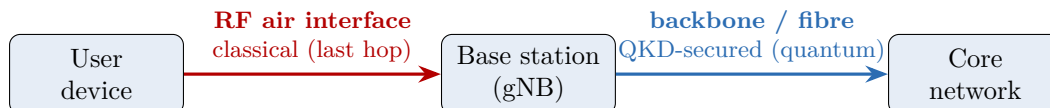
Shot noise is the irreducible blob size—the quantum noise floor you cannot beat (unlike thermal noise, which you can in principle cool away; Fig. 18). **CV-QKD** (the GG02 protocol) uses it for security: Alice draws X and P from a random *Gaussian* distribution and sends the corresponding coherent states; Bob homodyne-measures; their correlated numbers become a shared secret key. Any eavesdropper who measures the light *adds noise above the shot-noise floor* (and cannot clone it—no-cloning + $\Delta X \Delta P \geq \frac{1}{4}$), so she is detected.

“Is the modulation *really* Gaussian, or an approximation?” The Gaussian is a *design choice by Alice*, not a law of nature. She deliberately draws her (X, P) from a Gaussian because Gaussian modulation makes the security proof tight (Gaussian states are *extremal*: they give the eavesdropper the least advantage for a given power, so bounding her information is provably optimal). Nature does not force it—indeed *discrete-modulation* CV-QKD (a handful of fixed points, like a QPSK/QAM constellation) is an active, hardware-friendly alternative. So: the coherent states are physical; the *Gaussian cloud* is our chosen, provably-good way to arrange them.

Why this bridge matters to us. A CV quantum receiver *is* an I/Q receiver: the maths is the same X – P plane the team already works in on SDR [11, 12]. Our SDR path is therefore not a detour—it is the on-ramp to true quantum. Constellation design $\rightarrow$ Gaussian/discrete modulation; carrier/phase recovery $\rightarrow$ local-oscillator locking; SNR and BER $\rightarrow$ shot-noise floor and secret-key rate.

13. Quantum communication and 6G — overlay, not air interface

A mobile network has two very different segments. The **air interface** is the wireless RF hop between your device and the base station (the “last hop”). The **backbone** is the wired/fibre network linking base stations to the core. Quantum communication does *not* replace the RF air interface—as §8 explains, you cannot send single photons to a moving phone through walls (light is absorbed; only long-wavelength RF penetrates). Instead it is a **security overlay** (Fig. 19): quantum links (fibre + satellite QKD) secure the *backbone and key management*, while the RF last hop stays classical (protected by the keys quantum distributed).



Quantum secures the *backbone and keys*; the RF air interface stays classical. This is exactly where our *quantum-inspired* SDR physical-layer-security work fits.

Figure 19: Quantum communication as a security overlay on a mobile network: the RF air interface (last hop) stays classical, while quantum links (fibre + satellite QKD) secure the backbone and key management.

14. So what is *our* aim?

Our roadmap. We reach QSDC in four stages (Fig. 20), moving from the classical domain the team already commands toward the physical quantum domain:

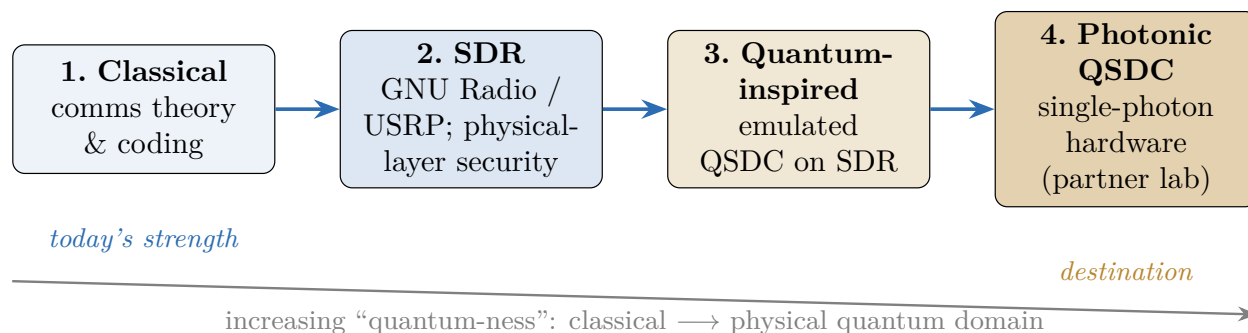


Figure 20: Nairi Quantum’s four-stage roadmap to QSDC, moving from the team’s classical/SDR strength (stages 1–2) through quantum-inspired emulation (stage 3) to photonic QSDC on single-photon hardware (stage 4).

Nairi Quantum works on Quantum Secure Direct Communication (QSDC)—quantum communication in which the *message itself*, not merely a key, is transmitted through a quantum (photonic) channel, secured by physics (no-cloning, entanglement, and $\Delta X \Delta P \geq \frac{1}{4}$) rather than computational hardness. The receiver is an engineering choice between DV (single-photon) and CV (coherent/homodyne, telecom- and SDR-friendly). Our roadmap begins in the classical/SDR domain—where the team is strongest and where the CV/I-Q bridge makes the transition natural—and advances toward true photonic QSDC with single-photon-capable partner hardware.

15. Glossary

Term	Meaning
Qubit	Two-level quantum system; $\alpha 0\rangle + \beta 1\rangle$. The unit of quantum information.
Superposition	A qubit being $ 0\rangle$ and $ 1\rangle$ at once until measured (not mere ignorance).
Measurement / collapse	Reading a qubit; forces a definite outcome and destroys the superposition.
No-cloning theorem	An unknown qubit cannot be copied — the basis of eavesdropper detection.
Entanglement	A non-classical correlation between qubits; a shared <i>resource</i> , not a message.
Bell state	A maximally entangled two-qubit state, e.g. $\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$.
QKD / QSDC	Distributes a <i>key</i> / sends the <i>message</i> through the quantum channel.
BB84	The first, most common QKD protocol; discrete-variable (single photons).
Teleportation	Transfers a qubit <i>state</i> using entanglement + 2 classical bits (not matter, not FTL).
SPDC	Spontaneous Parametric Down-Conversion — a crystal splits one photon into an entangled pair.
Quantum dot / NV centre	Solid-state single-photon sources (semiconductor dot / diamond defect).
Decoherence	Loss of a qubit's phase/superposition through coupling to the environment.
EM spectrum	Radio→microwave→IR→visible→UV→X-ray; one phenomenon, varied λ .
Optical / telecom light	Light in general / the low-loss fibre windows near 1310 & 1550 nm.
Cryogenic	Cooled to near absolute zero (needed for microwave-photon links).
DV / CV	Discrete-Variable (single-photon clicks) / Continuous-Variable (quadrature X, P).
Quadratures X, P	Amplitude & phase components of the field — the quantum I/Q.
Coherent state $ \alpha\rangle$	Ideal laser pulse; a minimum-uncertainty “blob” centred at $\alpha = X + iP$.
Coherence	A stable, well-defined phase of the wave (a laser has it; a bulb does not).
Shot noise	The irreducible quantum noise floor; $\Delta X = \Delta P = \frac{1}{2}$.
Homodyne detection	Measuring a quadrature by beating the signal against a local oscillator (I/Q-like).
OAM	Orbital Angular Momentum — a photon's helical phase twist; enables high-dimensional qudits.
Air interface / back-bone	Wireless last hop / wired-fibre core where quantum secures keys.

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